

PAST PAPER 2022

Q# 1:-

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Give one example for each of the categorical and Quantitative Variable. How would you enter in SPSS?

• Categorical Variable:-

Categorical Variable are those Variable that represent the name and label.

For Example:

- Marital status: (Single, Married)
- Smoking habit: (Smoker, non-smoker)

• Quantitative Variable:

Quantitative Variable are the Variable that represent the numerical values.

For example:

- Number of student in a class
- population size in a City.

click Transforms

Recode into different Variable

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Normal Q-Q-Plot of total Perceived anxiety.

It is the right tail Q-Q-Plot that shows the total Perceived cases of anxiety. It means that to the normal distribution there is little more data located at the extreme of the distribution and less data in the center of distribution.

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Box Plot:

A box plot is used to visualize the five number summary of a dataset which includes

The minimum : The first Quartile

The median : The third Quartile

The maximum:

□ Extreme outliers

The maximum number of elderly cohort is 85 and the extreme outlier is 85. The minimum value is 70. The third quartile is 78. The median is 75.

The first quartile is 73.

The range is

$$\begin{aligned}\text{Range} &= \text{maximum} - \text{minimum} \\ &= 85 - 70\end{aligned}$$

Interquartile range is

$$\text{maximum} - \text{minimum} / 2$$

~~div~~

Anxiety & Tension Crosstabulation.

This table shows the data of low and high anxiety and tension patients.

1. This is the row Percentage (i.e. out of the 26 who are low anxiety, 5 are low tension, $5/26 = 19.2\%$.)
2. This is the column Percentage (i.e. out of the 20 who are low tension)

5 one low anxiety, $5/20 = 25\%$.)

3. This is the cell %. (i.e. out of all 48, 5 are low anxiety, and low tension $5/48 = 10.4\%$.)

4. This is the overall row Percentage (i.e. out all 48, 26 were low anxiety, $26/48 = 54.27\%$.)

5. This is the overall column Percentage (i.e. out of all 48 people, 20 were low tension $20/48 = 41.7\%$.)

Histogram:-

it is a histogram which show that data is normally distributed because it's bell-shaped. The highest data is 21 and the lowest data is 3.

D#2:-

Enlist all the step to Test the Significant of Model

And regression Co-efficient:-

(i) $H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$

H_1 : at least one $\beta_i \neq 0$

(ii) Level of Significance

$$\alpha = 0.05$$

(iii) Test-statistics

$$F = \frac{\text{Reg MS}}{\text{Error MS}} \quad df_1 = n - k - 1$$

$$df_2 = k$$

(iv) We Will Reject H_0 if

P-value $< \alpha$

(v) Conclusion:-

Step of Regression Co-efficient.

$$H_0: \beta = 0$$

$$H_1: \beta \neq 0$$

$$\alpha = 0.05$$

P-value

Conclusion

R:-

R is the square root of R-Square and is the Correlation between the observed and the predicted

Value of dependent Variable

R-Square:-

R-Square is the Proportion of the Variance in the dependent Variable which can be Predicted from the independent Variable.

B:-

These are values for regression equation for Predicting the dependent Variable from the independent Variable. They are also called unstandardized Co-efficient because they are measured in their natural Units.

Ans:-

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\alpha = 0.05$$

$$t = \bar{x}_1$$

$$t = (\bar{x}_1 - \bar{x}_2)$$

$$SP \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

P-value

Write the P-value

Rejection Region

We will Reject H_0 if
 $P\text{-value} < \alpha$

Conclusion.

Highlighted steps:-

Mean:-

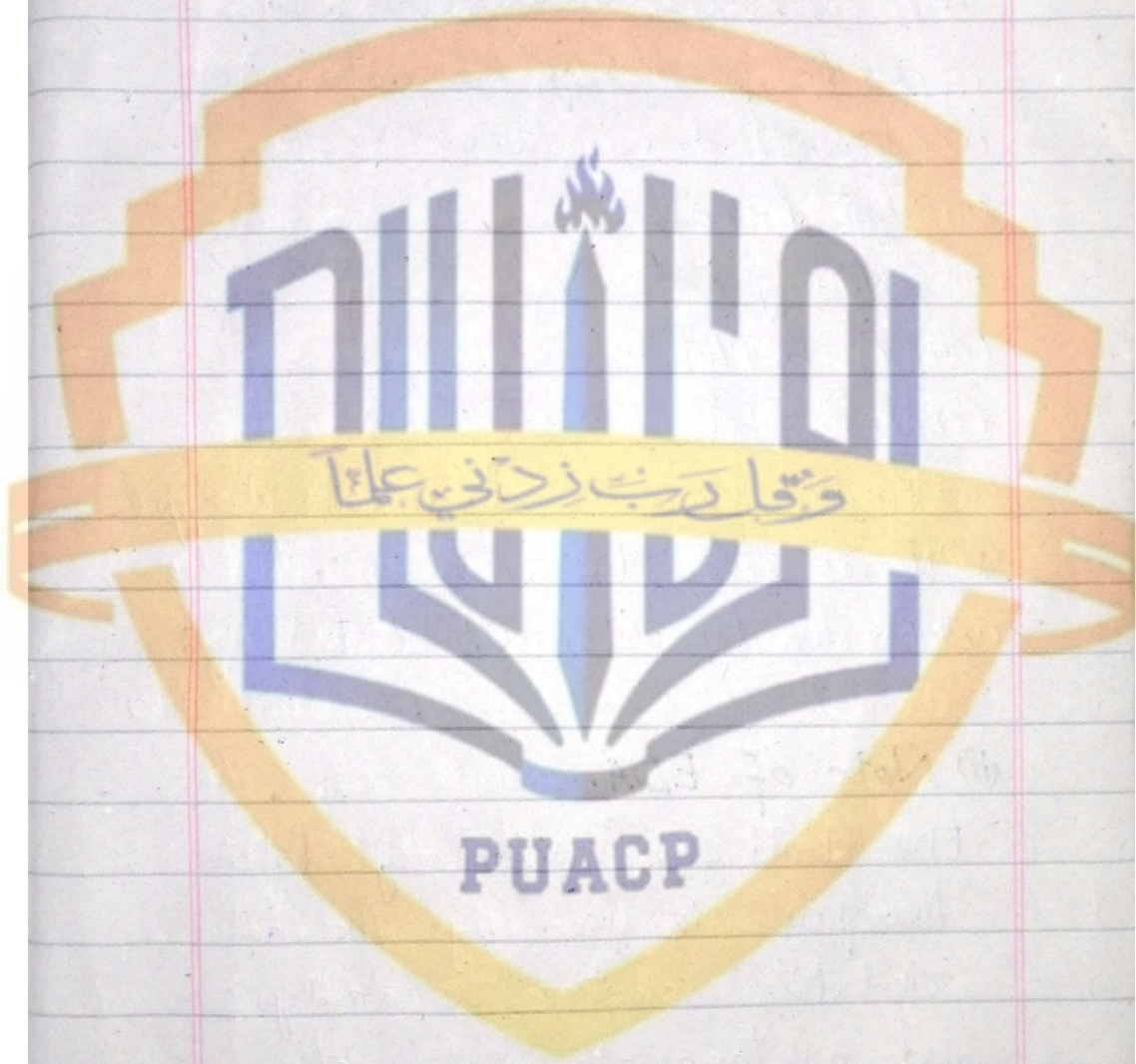
These are the respective mean
of the population Variable.

Correlation:-

This is the Correlation Coefficient
of the Pair of Variable indicated.
The correlation Co-efficient can
range from -1 to 1 , with
 -1 indicating a negative
Correlation, $+1$ indicating
the Positive Correlation and
 0 indicating no Correlation
at all.

Mean:-

This is the mean within
subject difference between
the two variable.



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